Department of Chemistry and
Department of Physics & Astronomy
University of Waterloo
200 University Avenue West
Waterloo, ON N2L 3G1, Canada
cgtetsas@uwaterloo.ca

August 29, 2026

The Editors
npj Materials Degradation
Nature Portfolio

Dear Editors,

We are pleased to submit our manuscript, **"Sparse data limit what a corrosion model can predict,"** for consideration as an Article in *npj Materials Degradation*.

Mechanistic corrosion models are routinely fitted with four to six parameters to a handful of measurements and then extrapolated across service lifetimes. Whether those parameters are determined by the data is almost never tested, and our central result is that, for a representative case, several of them are not, with consequences large enough to change a lifetime assessment. We establish this for the duplex oxide grown on Nb-stabilized AISI 347 (X6CrNiNb18-10) in simulated boiling-water-reactor hydrothermal water (240 °C, 7 MPa, 0.4 ppm dissolved O_2), using a differentiable, physics-informed inversion of the point defect model (PDM). It is well suited to *npj Materials Degradation*:

- **The first identifiability characterization of a PDM fit.** Profile likelihood, a 1000-member bootstrap, a prior-leverage test and a residual-leverage decomposition are built into the inversion rather than added afterward. They show that of the five parameters of the accepted model, three exposure times leave one undetermined and pin a second only to within an order of magnitude, and that 95% of the fit statistic is carried by three chromium points, and 80% by a single measurement. These tools are mature in adjacent fields but, to our knowledge, have not been brought to the PDM; the diagnosis transfers to any sparsely sampled degradation model.
- **Consequences large enough to matter, stated at the resolution the data support.** The identified kinetics fit the calibration data as well as an empirical power law but extrapolate differently: at ten years the power law lies a factor of 3.5 to 6.3 above the mechanistic barrier layer and outside its uncertainty band under every error treatment we apply, including one that widens that band from 447–704 nm to roughly 340–1040 nm. We are deliberate that the *separation* is the result and the point estimate it brackets (535 nm) is not: an analysis whose thesis is that sparse data under-determine parameters should not then quote a lifetime number as though it were determined. A designed exposure beyond roughly 3000 hours would resolve the separation, a concrete recommendation for lifetime assessment and the reason the re-analysis is worth doing on data already in the literature.
- **A diagnosed and remedied failure mode.** We document a physics-informed inversion failure specific to sparse experimental data, the data loss satisfied while the governing equation is not, and remedy it with a forward-solve self-consistency check, a cautionary and constructive result for the growing use of physics-informed learning in this field.
- **A negative result on the PDM's own constant-field closure.** Solving Poisson's equation self-consistently with the defect transport, instead of imposing the field, shows the closure

is not supported at the defect concentrations the model implies: the implied Debye length is about 0.06 nm, several times smaller than an interatomic spacing. This does not disturb the fitted kinetics, which identify b_3 rather than the field itself, but it bounds how the derived field strengths may be read, and we give it its own section rather than a footnote.

- **A first mechanistic description for this alloy, with its limits stated.** Prior characterization of AISI 347 oxide in BWR water stopped at empirical per-element power-law fits with no shared physical parameter set; we connect the Cr-rich barrier and Fe-rich outer layers through a single kinetic description. We are equally explicit that the model structure is not itself Nb-specific and that the neural inversion is validated against an exact classical reference rather than required by the problem.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. The author declares no competing interests. The work involves no human or animal subjects. All code, the digitized data, and a committed artefact behind every number in the manuscript are available in a public repository, together with a generated map from each figure and table to the command that produces it, so that any reported value can be re-derived without re-running the full pipeline. Thank you for considering our submission.

Sincerely,

Conrard Giresse Tetsassi Feugmo